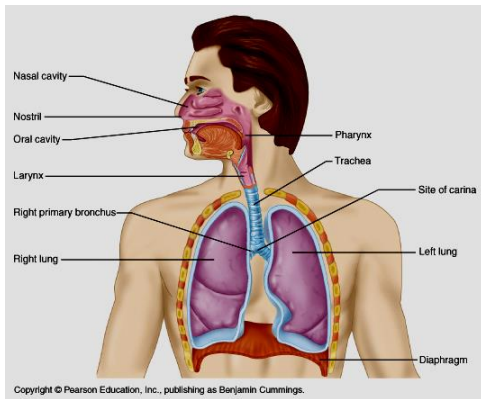




Respiratory system

- **Respiration**: exchange of O_2 / CO_2 between atmosphere, blood and cells
- **Cardiovascular** and **respiratory** systems work together to supply O_2 to and remove CO_2 from cells
- Provides exchange of O_2 and CO_2 to and from the blood
- **Organs**: nose, pharynx, larynx, trachea, bronchi, lungs

Functions of the Respiratory System

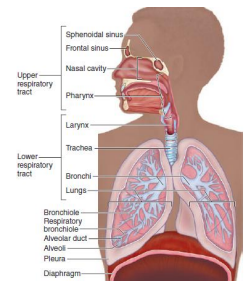
- **air conduction**
- Contains receptors for the sense of smell
- **filters** inspired air
- **gas exchange** (respiration)
- Sound production: air produces sounds
- Defense, Cough hairs, mucus, lymphoid tissue

Organization of the Respiratory System

➤ Structural (anatomical) classifications:

A. upper respiratory tract

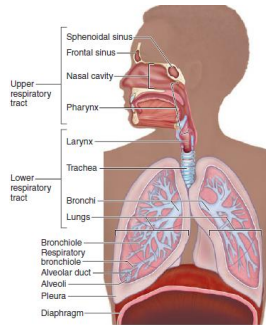
- Composed of
 - Nose
 - nasal cavity
 - paranasal sinuses
 - pharynx
 - and associated structures
- All are part of the **conducting** portion of the respiratory system



Cont....

B. lower respiratory tract

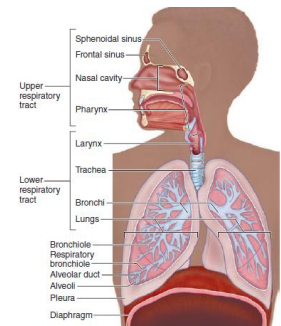
- Composed of:
- Larynx
- Trachea
- Bronchi
- Lungs



Cont....

➤ Functional classifications:**I. Conducting portion**

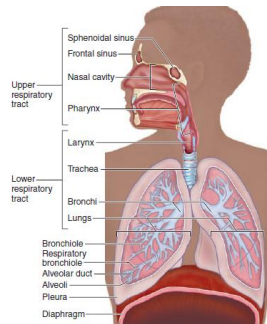
- transports air
- consists of:
 - nasal cavities
 - nasopharynx
 - larynx
 - trachea
 - bronchi
 - bronchioles,
 - and terminal bronchioles
- Upper respiratory tract is **all** conducting



Cont....

II. Respiratory portion

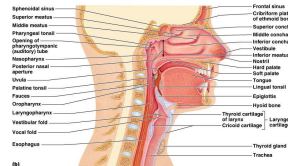
- where the system's **main** function of **gas exchange** occurs
- consisting of:
 - Respiratory bronchioles
 - alveolar ducts
 - and alveoli
- **Lower** respiratory tract has **both** conducting and respiratory portions.

**Nose**

- is part of the respiratory tract
- superior to the **hard palate** and contains the peripheral organ of smell
- functions of the nose:
 - **olfaction** (smelling)
 - **Respiration** (breathing)
 - **filtration** of dust
 - **humidification** of inspired air
 - reception and **elimination** of secretions from the **paranasal** sinuses and **nasolacrimal** ducts

Nasal Cavity

- Lies in and posterior to the **external** nose
- Is divided by a midline **nasal septum**
- Opens posteriorly into the nasal pharynx via **internal nares**
- The **ethmoid** and **sphenoid** bones form the roof



Nasal Cavity

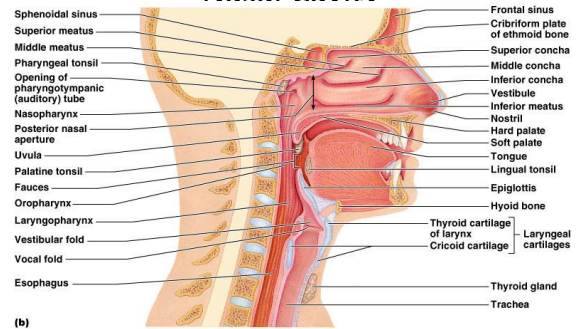
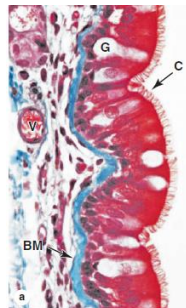


Figure 22.3b

Cont....

- The floor is formed by the hard and soft **palates**.
- lined by **Pseudostratified ciliated columnar** epithelial cells and **goblet** cells



Cont....

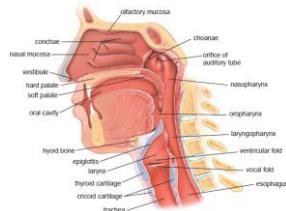
- divided into three regions
- A. Nasal Vestibule**
- a **dilated** space of the nasal cavity
- just inside nostrils and lined by **skin**
- contains hairs that **filter** coarse particles from inspired air



Cont....

B. Olfactory region

- is located at the apex (upper one third) of each nasal cavity
- lined by specialized **olfactory** mucosa
- Contains **smell** receptors



Cont....

C. Respiratory region

- the largest part (inferior two-thirds) of the nasal cavities
- lined by **respiratory** mucosa

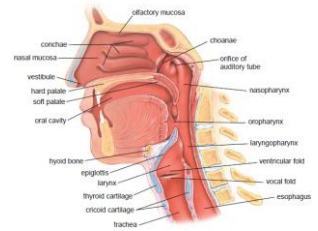
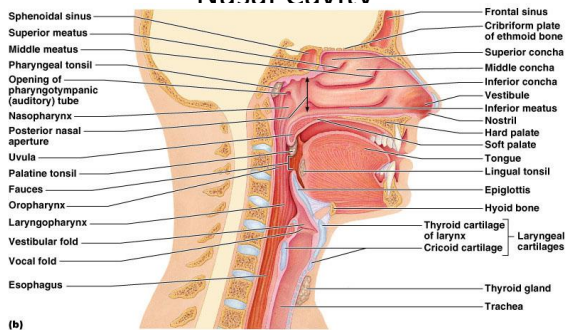
**Nasal Cavity**

Figure 22.3b

Cont...

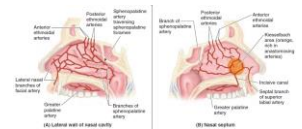
- Inspired air is:
 - Humidified by the high water content in the nasal cavity
 - Warmed by rich plexuses of capillaries
- Ciliated mucosal cells remove contaminated mucus
- **Superior, medial, and inferior** conchae:
 - Protrude medially from the lateral walls
 - Increase mucosal area
 - Enhance air turbulence and help filter air
- Sensitive mucosa triggers sneezing when stimulated by irritating particles

Functions of the Nasal Mucosa and Conchae

- During inhalation the conchae and nasal mucosa:
 - Filter, heat, and moisten air
- During exhalation these structures:
 - Reclaim heat and moisture
 - Minimize heat and moisture loss

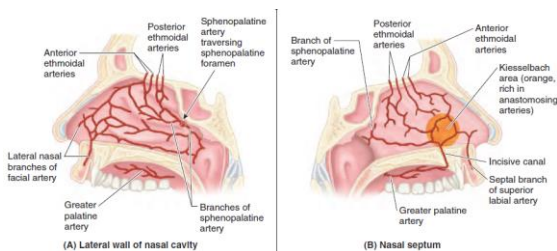
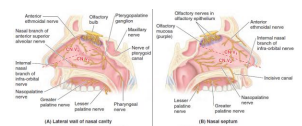
Vasculature and Innervation of Nose

- arterial supply of the medial and lateral walls of the nasal cavity is from five sources:
 1. Anterior ethmoidal artery
 2. Posterior ethmoidal artery
 3. Sphenopalatine artery
 4. Greater palatine artery
 5. Septal branch of the superior labial artery



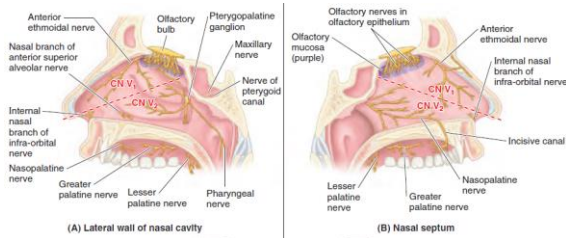
Cont....

1. Maxillary nerve
2. ophthalmic nerve
3. The **olfactory nerves**, concerned with smell, arise from cells in the **olfactory epithelium** in the superior part of the lateral and septal walls of the nasal cavity

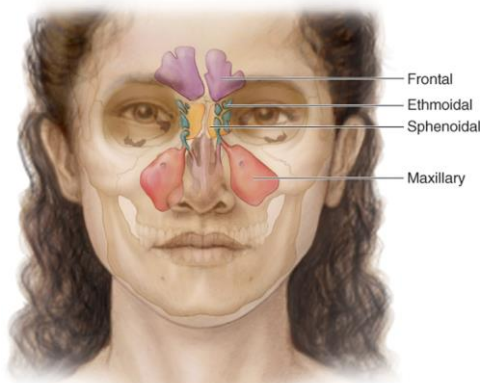


Paranasal Sinuses

- **Paranasal sinuses:**
 - In four skull bones
 - paired air spaces
 - **decrease** skull bone weight
- Named for the bones in which they are housed.
 - frontal
 - ethmoidal
 - Sphenoidal
 - maxillary
- Communicate with the nasal cavity by ducts
- Covered with the same **pseudostratified ciliated columnar epithelium** as the nasal cavity.

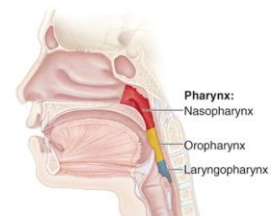


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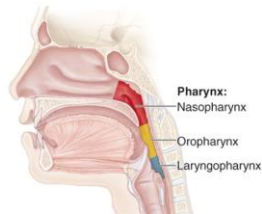
Pharynx

- Common to both the **respiratory** and **digestive** systems
- Commonly called the throat
- Funnel-shaped
- slightly wider superiorly and narrower inferiorly



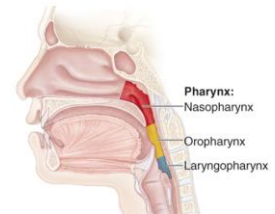
Cont....

- Originates **posterior** to the **nasal** and **oral** cavities
- Extends inferiorly near the level of the bifurcation of the **larynx** and **esophagus**.
- Common pathway for both **air** and **food**



Cont....

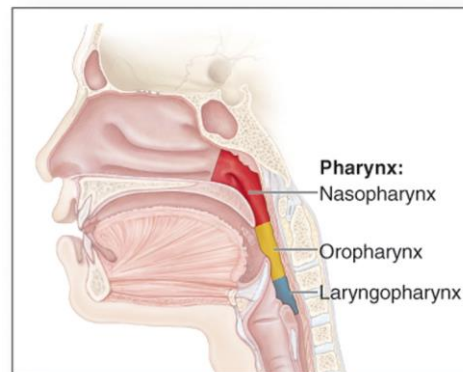
- Partitioned into three adjoining regions:
 - **Nasopharynx**
 - **Oropharynx**
 - **Laryngopharynx**



Cont...

- Walls:
 - lined by a mucosa
 - contain skeletal muscles primarily used for **swallowing**.
- Flexible lateral walls
 - distensible
 - to force swallowed food into the esophagus.

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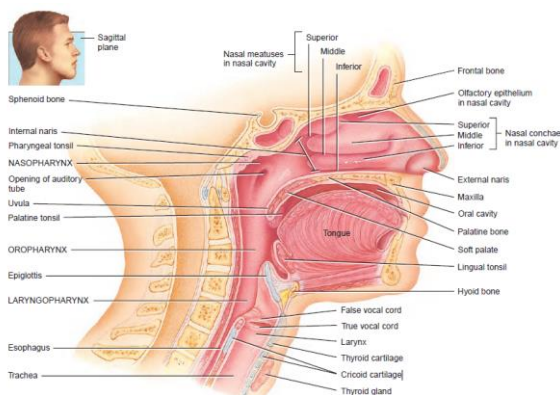
(b) Regions of pharynx

Nasopharynx

- Superiormost region of the pharynx.
- Location:
 - posterior to the nasal cavity
 - superior to the soft palate
 - separates it from the posterior part of the oral cavity.
- Normally, only air passes through.
- **Soft palate**
 - Blocks material from the oral cavity and oropharynx
 - elevates when we swallow.

Cont...

- **Auditory tubes**
 - paired
 - In the lateral walls of the nasopharynx
 - connect the nasopharynx to the **middle ear**.
- **Pharyngeal tonsil**
 - posterior nasopharynx wall
 - single
 - commonly called the **adenoids**.



Oropharynx

- The middle pharyngeal region.
- Location:
 - Immediately posterior to the oral cavity.
 - Bounded by the **soft palate** superiorly,
 - the **hyoid bone** inferiorly.
- Common respiratory and digestive pathway
 - both air and swallowed food and drink pass through.
- 2 pairs of muscular arches
 - **anterior palatoglossal arches**
 - **posterior palatopharyngeal arches**
 - form the entrance from the oral cavity.

Cont...

- Lymphatic organs
 - provide the “first line of defense” against ingested or inhaled foreign materials.
 - **Palatine tonsils**
 - on the lateral wall between the arches
 - **Lingual tonsils**
 - At the base of the tongue.

Laryngopharynx

- Inferior, narrowed region of the pharynx.
- Location:
 - Extends inferiorly from the **hyoid** bone
 - is continuous with the larynx and esophagus
 - Terminates at the **superior** border of the esophagus
 - is equivalent to the inferior border of the **cricoid** cartilage in the larynx.

Cont....

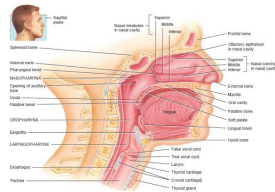
- The **larynx** (voice box) forms the anterior wall
- Lined with a **nonkeratinized stratified squamous epithelium** (mucus membrane)
- Permits passage of **both** food and air.

Lower Respiratory Tract

- **Conducting portion**
 - Larynx
 - Trachea
 - Bronchi
 - bronchioles and their associated structures
- **Respiratory portion** of the respiratory system
 - respiratory bronchioles
 - alveolar ducts
 - alveoli

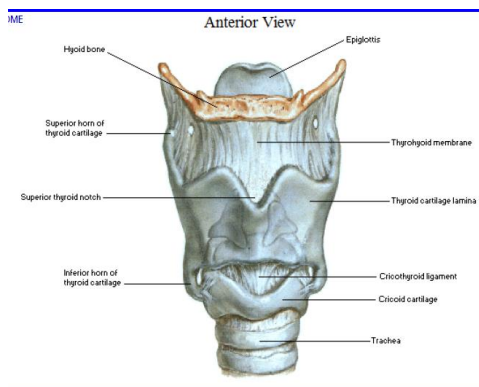
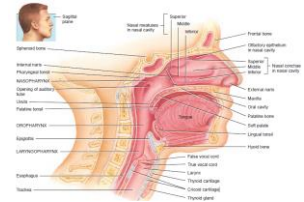
Larynx

- Short, somewhat cylindrical airway
- located in the anterior neck at the level of the bodies of C3–C6 vertebrae
- bounded posteriorly by the laryngopharynx,
- inferiorly by the trachea



Cont...

- Prevents swallowed materials from entering the lower respiratory tract
- Conducts air into the lower respiratory tract
- Produces sounds.



Cont...

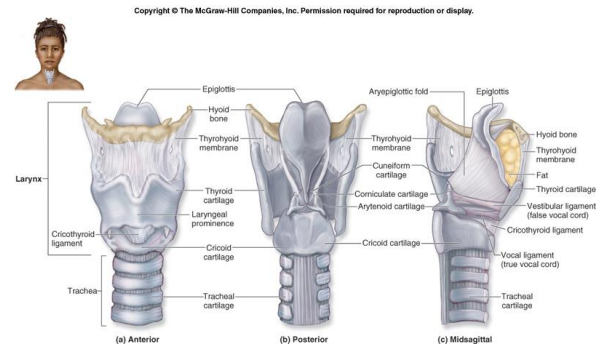
■ Laryngeal Skeleton:

- consists of **nine** cartilages.

-Thyroid	} Single
-Cricoid	
-and epiglottic	
-Arytenoid	} Paired
-Corniculate	
and cuneiform	

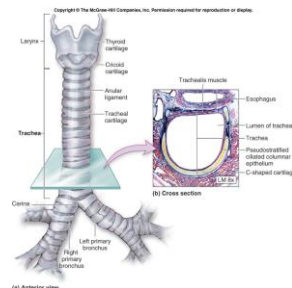
Cont...

- held in place by ligaments and muscles.
- Intrinsic muscles: regulate tension on true vocal cords
- Extrinsic muscles: stabilize the larynx



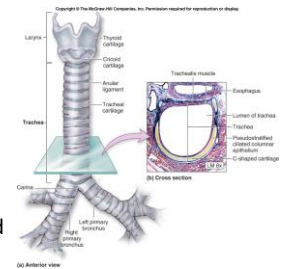
Trachea

- is a short, flexible, air tube
- **2.5 cm** in diameter and **10 cm** long
- **anterior** to the esophagus
- extend from the **larynx** into the thorax
- terminates inferiorly as it divides into right and left **main bronchi**

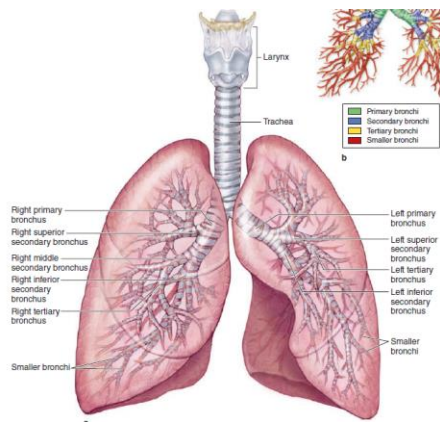


Cont....

- Anterior and lateral walls of the trachea are supported by **15 to 20 C-shaped tracheal cartilages**
- cartilage rings reinforce and provide some rigidity to the tracheal wall to ensure that the trachea remains **open (patent)** at all times
- trachealis muscle, smooth muscle connect the ends of the rings
- cartilage rings are connected by elastic sheets called **anular ligaments**.



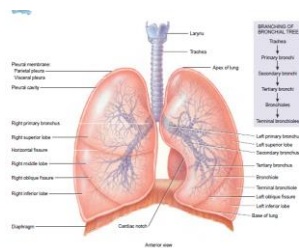
Cont...



- Pseudostratified ciliated columnar epithelium
- At the level of the **sternal angle**, the trachea bifurcates into two smaller tubes, called the **right and left primary bronchi**
- Each primary bronchus projects laterally toward each lung
- The most inferior tracheal cartilage separates the primary bronchi at their origin and forms an internal ridge called the **carina**

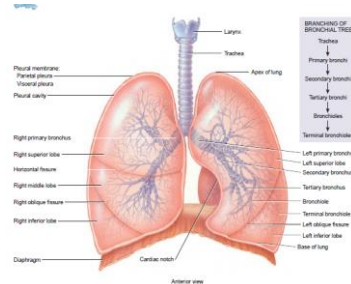
Bronchial Tree

- A highly branched system
 - air-conducting passages
 - originate from the left and right **primary bronchi**
- Progressively branch into narrower tubes as they diverge throughout the lungs before terminating in **terminal bronchioles**
- Consists of airways that begin at the trachea and end at the **terminal bronchioles**.



I. Primary bronchi

- Incomplete rings of hyaline cartilage ensure that they remain open



Cont...

Cont...

- Right primary bronchus
 - shorter, wider, and more vertically oriented than the left primary bronchus
 - Foreign particles are more likely to lodge in the right primary bronchus.
- **Primary bronchi**
 - enter the hilum of each lung
 - Also entering hilum:
 - pulmonary vessels
 - lymphatic vessels
 - nerves.

Cont....

II. Secondary bronchi (or lobar bronchi)

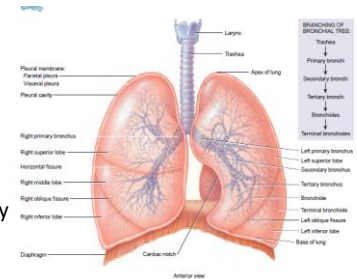
- Branch of primary bronchus

○ left lung:

- two lobes
- two secondary bronchi

○ right lung

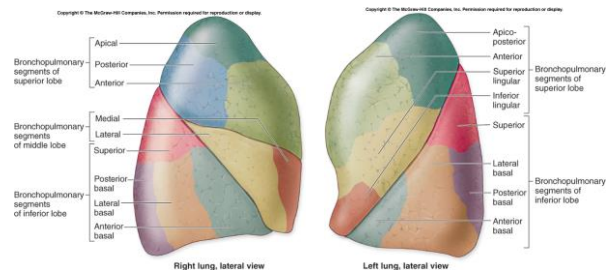
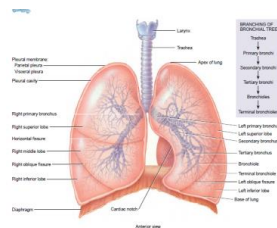
- three lobes
- three secondary bronchi



Cont....

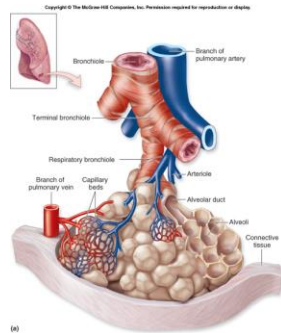
III. Tertiary bronchi (or segmental bronchi)

- Branch of secondary bronchi
- left lung is supplied by **8 to 10** tertiary bronchi
- right lung is supplied by **10** tertiary bronchi
- supply a part of the lung called a **bronchopulmonary segment**



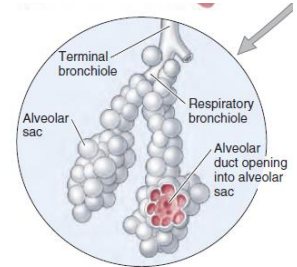
Respiratory Bronchioles, Alveolar Ducts, and Alveoli

- Each **terminal bronchiole** gives rise to several generations of **respiratory bronchioles**
- characterized by scattered, thin-walled **outpocketings (alveoli)** that **extend** from their **lumens**
- involved both in air **transportation** and **gas exchange**



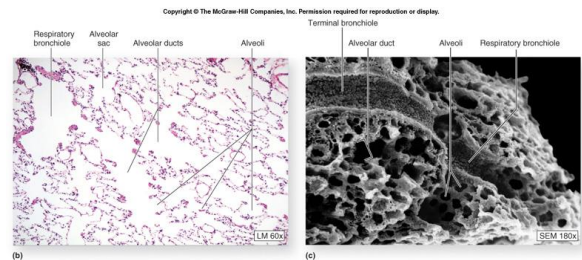
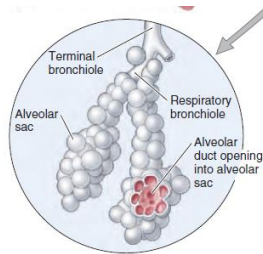
Cont....

- Each respiratory bronchiole gives rise to 2–11 **alveolar ducts**, each of which gives rise to 5–6 **alveolar sacs**
- Alveolar ducts** are elongated airways **densely** lined with alveoli, leading to common spaces, the **alveolar sacs**, into which clusters of alveoli open



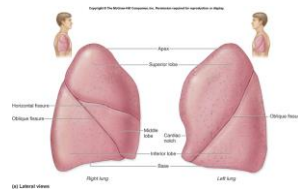
Cont....

- An alveolus is about **0.25** to **0.5** millimeter in diameter
- Its thin wall is specialized to **promote diffusion of gases** between the alveolus and the blood in the **pulmonary capillaries**
- Gas exchange can take place in the **respiratory bronchioles** and **alveolar ducts** as well as in the **lungs**, which contain approximately **300–400 million alveoli**
- The **spongy** nature of the lung is due to the packing of millions of alveoli together

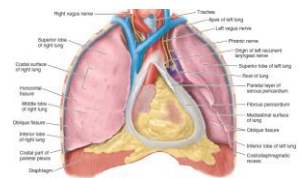


Gross Anatomy of the Lungs

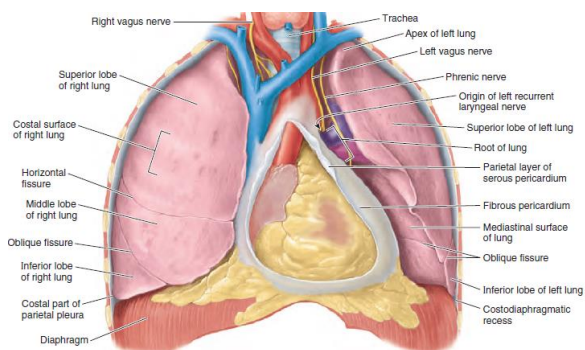
- Each lung has a **conical** shape
- Its wide, concave **base** rests upon the **muscular diaphragm**.
- Its relatively blunt superior region, called the **apex** or (cupola), projects superiorly to a point that is slightly superior and posterior to the clavicle.



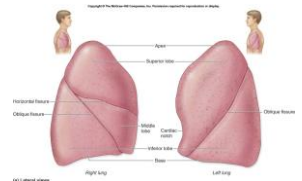
- Both lungs are bordered by the thoracic wall anteriorly, laterally, and posteriorly, and supported by the **rib cage**.
- Toward the midline, the lungs are separated from each other by the **mediastinum**.
- The relatively broad, rounded surface in contact with the thoracic wall is called the **costal surface** of the lung.



Cont....



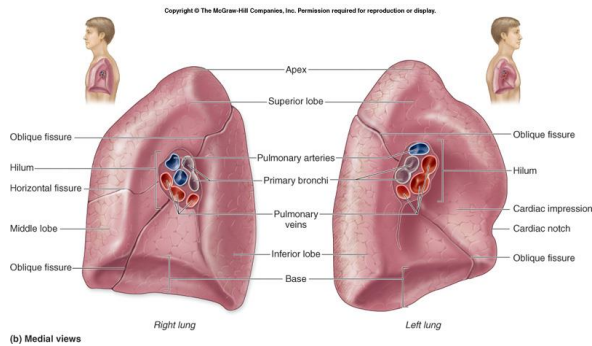
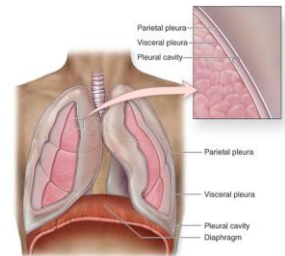
- The left lung
 - has the cardiac notch.
 - is 10% smaller than the right lung.
 - has two lobes (**superior** and **inferior** lobes) divides by the **oblique** fissure.
- ✓ The **oblique** and **horizontal** fissures divide the right lung into **superior**, **middle**, and **inferior** lobes



Cont....

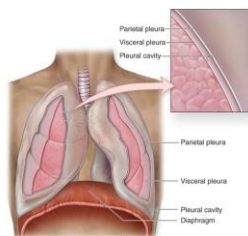
Pleura and Pleural Cavities

- The outer surface of each lung and the adjacent internal thoracic wall are lined by a serous membrane called **pleura**, which is formed from **simple squamous epithelium**
- The **outer** surface of each **lung** is tightly covered by the **visceral pleura**



Cont....

- the **internal thoracic** walls, the **lateral** surfaces of the **mediastinum**, and the **superior** surface of the **diaphragm** are lined by the **parietal pleura**.
- The parietal and visceral pleural layers are continuous at the **hilum** of each lung.



Cont....

- The **potential space** between these serous membrane layers is a **pleural cavity**
- The pleural membranes produce a thin, serous **fluid** that circulates in the pleural cavity and acts as a **lubricant**, ensuring **minimal friction** during breathing.

